STANDARD OPERATING PROCEDURE
VPN Connectivity Troubleshooting — Remote Employees

SOP Number:	SOP-IT-VPN-001
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Owner:	IT Operations — Network & Security Team
Applicability:	All remote employees accessing corporate resources via VPN
Classification:	Confidential — Internal Use Only

1. Purpose

This Standard Operating Procedure (SOP) establishes a uniform, repeatable process for diagnosing and resolving Virtual Private Network (VPN) connectivity failures affecting remote employees. Adherence to this procedure ensures consistent service restoration, accurate incident documentation, and appropriate escalation to Tier 2 and Tier 3 support where required.

2. Scope

This procedure applies to all remote employees and contractors experiencing VPN connectivity issues on corporate-managed and BYOD devices. It covers the following supported platforms:

- Windows 10 / Windows 11 (GlobalProtect Agent v6.x)
- macOS 12 Monterey and later (GlobalProtect Agent v6.x)
- Ubuntu 20.04 LTS / 22.04 LTS (OpenConnect CLI)
- iOS 15+ / Android 12+ (GlobalProtect Mobile)

3. Definitions

Term	Definition
VPN	Virtual Private Network — encrypted tunnel between the remote endpoint and corporate gateway.
GlobalProtect	Palo Alto Networks VPN client deployed enterprise-wide.
MFA	Multi-Factor Authentication — required for all VPN sessions via Duo Security.
PAN-GP Gateway	Corporate VPN gateway hosted at vpn.corp.example.com (Primary) and vpn-dr.corp.example.com (DR).

4. Common Error Codes & Initial Actions

Error Code	Description	Initial Corrective Action
`GP-0001`	Authentication failure — invalid credentials	Reset AD password via self-service portal; verify Duo push accepted.
`GP-0007`	Certificate validation failed	Re-enroll device certificate via SCEP; restart GP service.
`GP-0013`	Gateway unreachable / timeout	Verify internet connectivity; attempt DR gateway.

Error Code	Description	Initial Corrective Action
`GP-0021`	HIP check failure — non-compliant endpoint	Run Windows Update / apply pending patches; confirm AV definitions current.
`GP-0034`	Split-tunnel route conflict	Flush routing table; see Section 5, Step 4.
`ERR-AUTH-403`	MFA token expired or denied	Re-initiate Duo authentication; check device clock sync (NTP).
`ERR-TLS-SSL`	TLS handshake error	Verify system clock accuracy; check proxy interference.
`ERR-DNS-NXDM`	DNS resolution failure post-connect	Flush DNS cache; see Section 5, Step 6.

5. Step-by-Step Troubleshooting Procedure

Step 1 — Verify Basic Internet Connectivity

Before engaging VPN diagnostics, confirm the endpoint has active internet access.

Windows / macOS:

```
ping -c 4 8.8.8.8
traceroute vpn.corp.example.com
```

Linux:

```
ping -c 4 1.1.1.1 && mtr --report vpn.corp.example.com
```

Expected result: Round-trip time < 200 ms with no packet loss. If connectivity is absent, troubleshoot local ISP or network before proceeding.

Step 2 — Confirm GlobalProtect Agent Version

Outdated agent versions are a common source of authentication and HIP check failures (GP-0021).

Windows (PowerShell — Run as Administrator):

```
Get-ItemProperty 'HKLM:\SOFTWARE\Palo Alto Networks\GlobalProtect' | Select-Object
Version
```

macOS / Linux:

```
globalprotect --version
```

Minimum required version: 6.2.4. If below minimum, download the current installer from the IT Self-Service Portal at https://portal.corp.example.com/vpn.

Step 3 — Collect and Review Diagnostic Logs

Capture GP logs before any configuration changes.

Windows:

```
C:\Program Files\Palo Alto Networks\GlobalProtect\pangpd.log
Get-Content "$env:APPDATA\Palo Alto Networks\GlobalProtect\PanGPA.log" -Tail 100
```

macOS:

```
tail -100 /Library/Application\ Support/PaloAltoNetworks/GlobalProtect/pangpd.log
```

Linux:

```
journalctl -u openconnect --since '1 hour ago' | tail -80
```

Attach relevant log excerpts to the helpdesk ticket before escalation.

Step 4 — Flush Routing Table (Error GP-0034)

Split-tunnel conflicts prevent corporate traffic from routing correctly through the VPN interface.

Windows (Administrator PowerShell):

```
route delete 0.0.0.0
route add 0.0.0.0 mask 0.0.0.0 <default_gateway_IP>
ipconfig /flushdns && ipconfig /release && ipconfig /renew
```

macOS / Linux:

```
sudo ip route flush cache
```

```
sudo resolvectl flush-caches   # systemd-resolved
sudo dscacheutil -flushcache && sudo killall -HUP mDNSResponder   # macOS
```

Step 5 — Restart GlobalProtect Service

Windows (Administrator PowerShell):

```
Stop-Service -Name 'PanGPS' -Force; Start-Service -Name 'PanGPS'
Get-Service PanGPS | Select-Object Status
```

macOS:

```
sudo launchctl stop com.paloaltonetworks.globalprotect
sudo launchctl start com.paloaltonetworks.globalprotect
```

Linux (OpenConnect):

```
sudo systemctl restart openconnect
sudo systemctl status openconnect
```

Step 6 — DNS Resolution Verification Post-Connection (ERR-DNS-NXDM)

Confirm VPN DNS servers are active after tunnel establishment.

Windows:

```
nslookup intranet.corp.example.com 10.0.1.53
Resolve-DnsName intranet.corp.example.com -Server 10.0.1.53
```

macOS / Linux:

```
dig @10.0.1.53 intranet.corp.example.com
```

Expected result: IP address returned from the 10.x.x.x corporate subnet. If NXDOMAIN is returned, escalate to Tier 2 — DNS/AD Team.

6. Escalation Protocols

If the steps outlined in Section 5 do not resolve the issue within 30 minutes, the matter must be formally escalated in accordance with the following tiered protocol.

6.1 Tier 1 — Service Desk (Initial Response)

Responsible Team: IT Service Desk

SLA: Acknowledgement within 15 minutes; resolution attempt within 1 hour.

Actions at Tier 1:

- Log incident in ServiceNow with error code, OS/agent version, and timestamp.
- Execute all steps in Section 5 in sequence.
- Attempt connection to DR gateway: vpn-dr.corp.example.com.
- Document outcomes in ticket before escalation.

Escalate to Tier 2 if: error persists after full Section 5 execution, multiple users report concurrent failures (possible gateway outage), or error code is GP-0007 (certificate) or GP-0021 (HIP) with confirmed compliance.

6.2 Tier 2 — Network & Security Engineering

Responsible Team: Network Engineering — #net-sec-vpn (Slack)

SLA: Acknowledgement within 30 minutes during business hours; 1 hour off-hours.

Actions at Tier 2:

- Review PAN-OS gateway logs via Panorama dashboard.
- Validate HIP match policy for affected device profile.
- Inspect gateway authentication logs for credential or certificate anomalies.
- Issue emergency certificate re-enrollment via SCEP if GP-0007 persists.
- Check Duo Security admin console for MFA policy exceptions or failed push logs.

Escalate to Tier 3 if: gateway infrastructure fault is confirmed, IPSec/SSL parameter mismatch is detected, or incident affects >10 users simultaneously.

6.3 Tier 3 — Security Operations / Vendor Support

Responsible Team: Security Operations Center (SOC) + Palo Alto Networks TAC

SLA: Engagement within 60 minutes; Palo Alto Networks TAC case opened within 2 hours.

Actions at Tier 3:

- Declare VPN P1 incident via PagerDuty if >20 users are impacted.
- Initiate failover to DR gateway cluster per Runbook DR-NET-001.
- Open Palo Alto Networks TAC case at support.paloaltonetworks.com with full Panorama tech-support file.
- Notify IT Director and CISO via incident bridge call.

7. Escalation Contact Matrix

Tier	Team	Contact Channel	After-Hours
Tier 1	IT Service Desk	ext. 4357 / helpdesk@corp.example.com	On-call rotation via PagerDuty
Tier 2	Network & Security Engineering	#net-sec-vpn (Slack) / ext. 4210	PagerDuty: Network-Oncall policy
Tier 3	SOC + PAN TAC	soc@corp.example.com / PAN TAC: 1-844-PAN-0100	PagerDuty: SOC-P1 policy; notify IT Director

8. Documentation & Ticket Requirements

All VPN incidents must be logged in ServiceNow. Each ticket must contain the following fields at minimum:

- Incident timestamp (UTC), user UPN, and device hostname.
- GlobalProtect agent version and OS build number.
- Exact error code(s) displayed or retrieved from logs.
- Steps performed (reference section numbers from this SOP).
- Outcome of each step (resolved / unresolved / partial).
- Escalation tier engaged, if applicable, and time of escalation.

Tickets must remain open until the root cause has been confirmed and documented. Do not close incidents on the basis of symptom resolution alone without root cause identification.

9. Revision History

Rev.	Date	Author	Summary of Changes
1.0	03 Jan 2025	J. Thornton, IT Operations	Initial release.
2.0	15 Sep 2025	M. Kapoor, Network Engineering	Added GP-0034 routing fix; updated escalation contacts.
2.1	**01 May 2026**	**S. Okafor, Security Operations**	**Added ERR-AUTH-403, ERR-TLS-SSL, ERR-DNS-NXDM; Linux commands; updated MFA to Duo.**